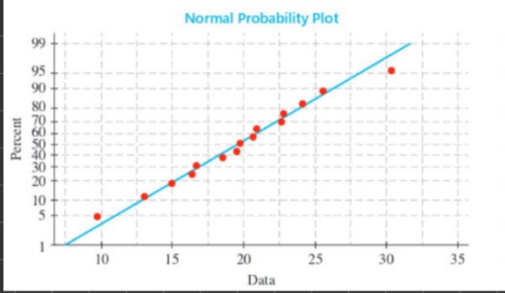
Is the Data Normal? *Use the graphical displays in Exercises 12–15 to decide whether the data have been selected from a normal population. Explain your answer.*

13.



Based on the normal probability plot in Exercise 13, the data appear to come from a population that is normally distributed. This is because the data points fall approximately along a straight line, showing no significant departures or patterns that would suggest a non-normal distribution.

20. Cerebral Blood Flow Cerebral blood flow (CBF) in the brains of healthy people is normally distributed with a mean of 74. A random sample of 25 stroke patients resulted in an average CBF of $\bar{x} = 69.7$ with a standard deviation of $s = 16$. Test the hypothesis that the average CBF for stroke patients is different from that of healthy people using $\alpha = .05$.

$\neq$

$$\star \quad t = \frac{\bar{x} - \mu_0}{s\sqrt{n}}$$

$\mu_0 = 74$ $\quad n = 25$

$\bar{x} = 69.7$ $\quad \alpha = 0.05$

$s = 16$ $\quad df = n-1 = 24$

$$t = \frac{69.7 - 74}{16/\sqrt{25}} = \frac{-4.3}{16/5} = \frac{-4.3}{3.2} = -1.34375 \approx -1.34$$

$$\alpha = 0.05 \div 2 = 0.025$$

• 臨界值 $t_{\alpha/2} \approx \pm 2.064$

t值未達拒絕域，無法拒絕 H_0

30. Bulimia An article in the *British Journal of Clinical Psychology* indicates that self-esteem was an important predictor of those who will benefit from treatment for bulimia.[4] The table gives the mean and standard deviation of self-esteem scores prior to treatment, at posttreatment, and during a follow-up:

	Pretreatment	Posttreatment	Follow-up
Sample Mean $\bar{x}$	20.3	26.6	27.7
Standard Deviation s	5.0	7.4	8.2
Sample Size n	21	21	20

a. Use a test of hypothesis to determine whether there is sufficient evidence to conclude that the true pretreatment mean is less than 25.

b. Construct a 95% confidence interval for the true posttreatment mean.

c. In Section 10.3, we will introduce small-sample techniques for making inferences about the difference between two population means. Without a formal statistical test, what might you conclude about the differences among the three sampled population means represented by the results in the table?

a. $\mu = 25$, $\bar{x} = 20.3$, $s = 5.0$, $n = 21$

$$t = \frac{20.3 - 25}{5.0/\sqrt{21}} = \frac{-4.7}{1.091} \approx -4.31$$

$df = n - 1 = 21 - 1 = 20$

$\alpha = 0.05$, $t_{0.05} \approx -1.725$

$\because -4.31 < -1.725$

$\therefore$ 自尊的平均得分低於 25

b. $\bar{x} = 26.6$, $s = 7.4$, $n = 21$

$df = 21 - 1 = 20$ $\quad 0.5 \div 2 = 0.025$

$t_{0.025} \approx 2.086$

$\frac{7.4}{\sqrt{21}} = 1.615$

$E = 2.086 \cdot \frac{7.4}{\sqrt{21}} = 2.086 \cdot 1.615 \approx 3.37$

$26.6 \pm 3.37 \Rightarrow [23.23, 29.97]$

$\therefore$ 自尊的平均得分落在 23.23 到 29.97 之間

☆ $\bar{x} \pm t_{\alpha/2} \cdot \frac{s}{\sqrt{n}}$

$SE = \frac{s}{\sqrt{n}}$

$ME = t \times SE$

c. Means → pre 20.3

→ post 26.6

→ follow up 27.7

$\therefore$ post > pre, follow up stays high,

post and follow-up likely similar

35. Sodium Chloride Measurements of water intake, obtained from a sample of 17 rats that had been injected with a sodium chloride solution, produced a mean and standard deviation of 31.0 and 6.2 cubic centimeters (cm^3), respectively. Suppose that the average water intake for noninjected rats observed over a comparable period of time is 22.0 cm^3.

a. Do the data indicate that injected rats drink more water than noninjected rats? Test at the 5% level of significance.

b. Find a 90% confidence interval for the mean water intake for injected rats.

☆ $\bar{x} \pm t_{\alpha/2} \cdot \frac{s}{\sqrt{n}}$

$\mu_0 = 22$, $\bar{x} = 31$, $s = 6.2$

$n = 17$, $df = n-1 = 17-1 = 16$

a. $t = \frac{31.0 - 22.0}{6.2/\sqrt{17}} = \frac{9.0}{1.503} \approx 5.988$

$df = 16$, $\alpha = 0.05$, $t_{0.05} \approx 1.746$

$\because t = 5.988 > 1.746$

$\therefore$ 注射食鹽水的老鼠 > 未注射的老鼠

b. $df = 16$, $t_{0.05} \approx 1.746$

$E = 1.746 \cdot \frac{6.2}{\sqrt{17}} = 1.746 \cdot 1.503 \approx 2.62$

$31.0 \pm 2.62 \Rightarrow (28.38, 33.62)$

題目情境	關鍵字	查表欄位 (α 或 $\alpha/2$)
單尾檢定 ($\alpha = 0.05$)	$>$, $<$, More, Less	0.05
雙尾檢定 ($\alpha = 0.05$)	$\neq$, Different	0.025
90% 信賴區間	Confidence Interval	0.05
95% 信賴區間	Confidence Interval	0.025
99% 信賴區間	Confidence Interval	0.005

Two-Sample Tests *Use the information provided in Exercises 7–8. State the null and alternative hypotheses for testing for a significant difference in means, calculate the pooled estimate of σ^2, the associated degrees of freedom, and the observed value of the t statistic. What is the rejection region using $\alpha = .05$? What is the p-value for the test? What can you conclude from these data?*

7.

Population 1	12	3	8	5		→ 4
Population 2	14	7	7	9	6	→ 5

- $\mu_1 - \mu_2 \neq 0$

$df = n_1 + n_2 - 2 = 4 + 5 - 2 = 7$

$$S_p^2 = \frac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{n_1+n_2-2} = \frac{46 + 41.2}{7} = \frac{87.2}{7} \approx 12.457$$

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{S_p^2\left(\frac{1}{n_1}+\frac{1}{n_2}\right)}} = \frac{12.0 - 8.6}{\sqrt{12.457\left(\frac{1}{4}+\frac{1}{5}\right)}} = \frac{-1.6}{\sqrt{12.457 \times 0.45}} = \frac{-1.6}{2.367} \approx -0.676$$

$\alpha = 0.05$. $df = 7$. 臨界值 ± 2.365

拒絕域 $t > 2.365$, $t < -2.365$

$t = -0.676$, p-value > 0.05